EXPT-9

17-09-26

IMPLEMENTATION OF SUBNETTING IN CISCO PACKET TRACER.

Aim:-

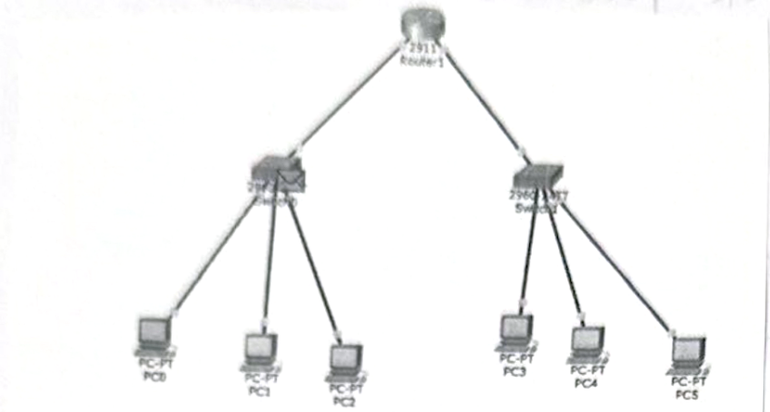
To implement subnetting using Cisco Packet Tracer, assign static IP addressing, connect different subnets using a router, and observe packet transmission.

THEORY:-

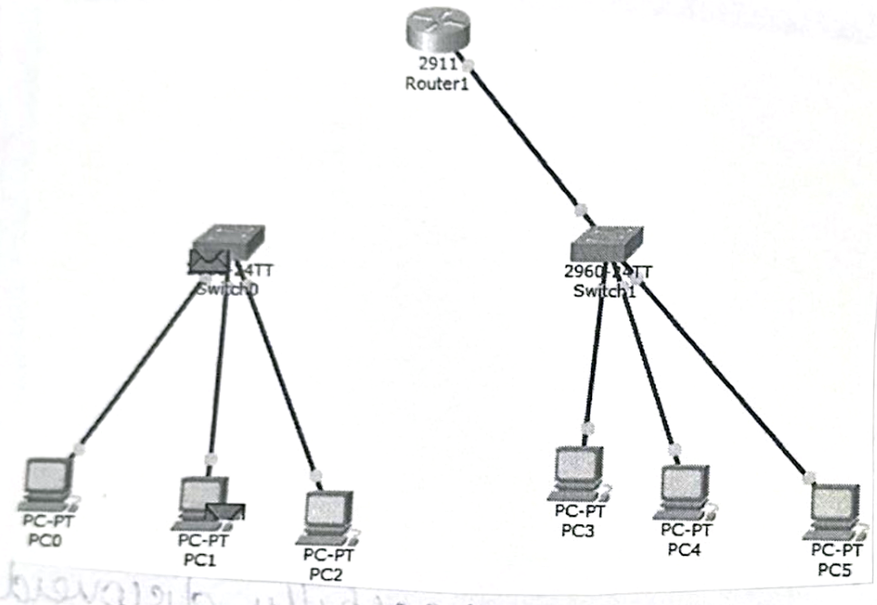
Subnetting divides a network into smaller networks called subnets. It helps in efficient IP address usage and allows devices to be organised into separate networks.

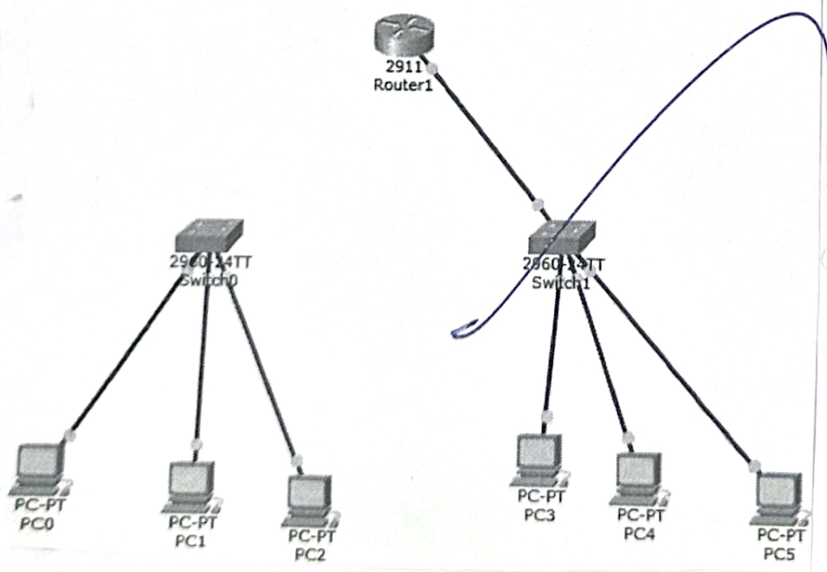
PROCEDURE:-

1. Create the Topology:-
   - Open the Cisco Packet Tracer.
   - Add routers, switches and PCs.
   - Connect the devices using appropriate cables
2. Perform subnetting:
   - Use the network 192.168.1.0/24.
   - Divide it into smaller subnets using /27.
   - /27 provides 8 subnets, with 30 usable host addresses in each subnet.



Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
Successful	PC0	PC3	ICMP		0.000	N	0	(edit)	





3. Assign IP address.

↳ Assign different subnet IPs to the PCs.

↳ Assign the first usable IP of each subnet to the routes interface as the default gateway

4. Configure the router:

```
enable
configure terminal
interface FastEthernet 0/0
ip address 192.168.1.1 255.255.255.224.
no shutdown.
interface FastEthernet 0/1.
ip address 192.168.2.1 255.255.255.224
no shutdown.
```

5. Configure the switch:

↳ Enter the switch CLI.

↳ Configure the required interfaces as access ports.

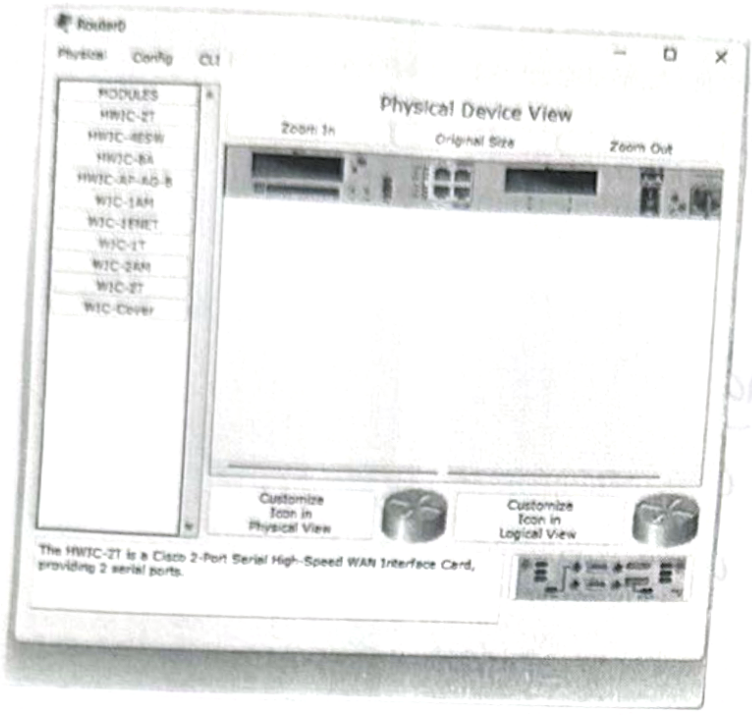
```
enable
configure terminal.
interface fastEthernet0/1
switchport mode access.
```

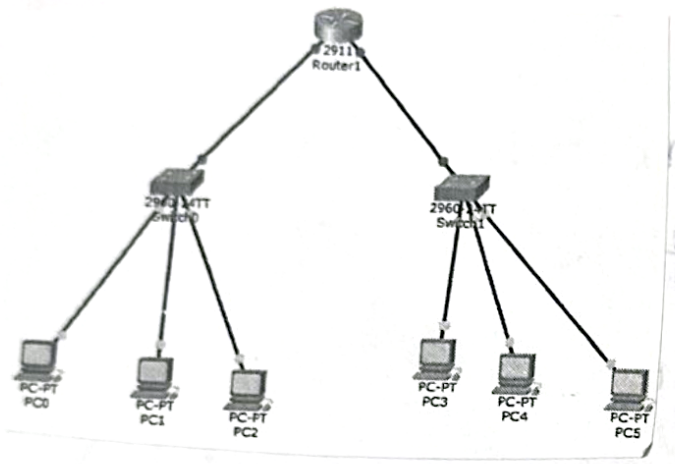
6. Test Communication:-

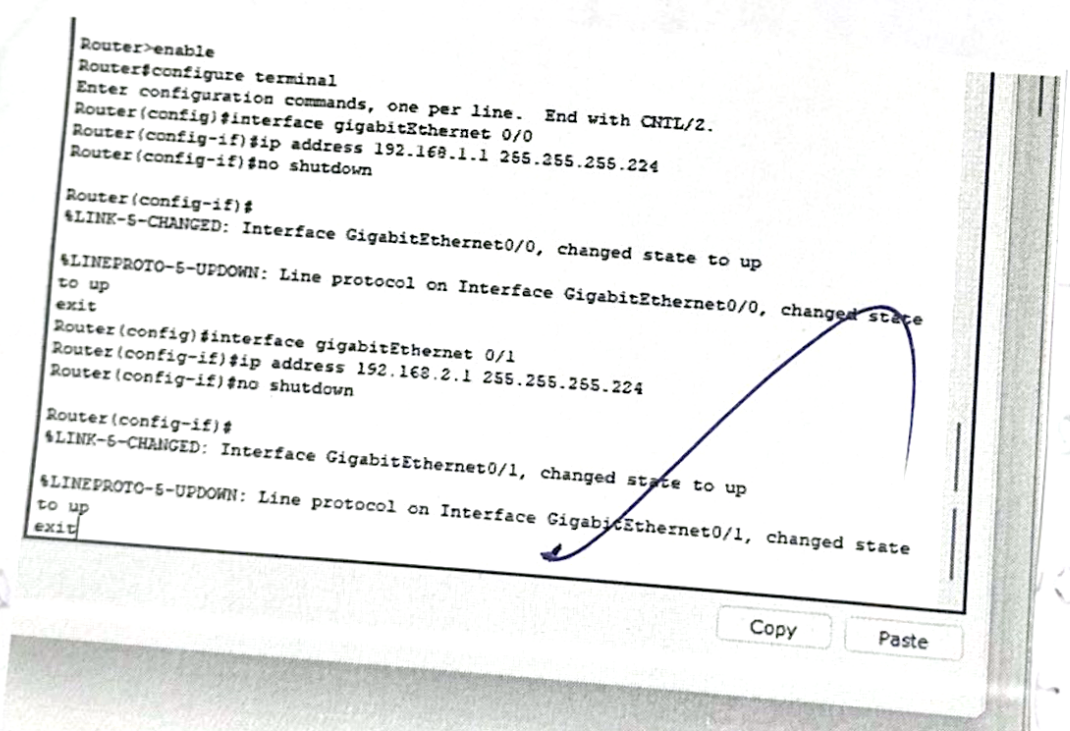
↳ Open the command prompt.

↳ Use ping to test communication between PCs.

↳ Test both PCs in the same subnet and PCs belonging to different subnets.


Router0
Physical
Config
CLI
MODULES
HWIC-2T
HWIC-4ESW
HWIC-8A
HWIC-AP-AG-B
WIC-1AM
WIC-1ENET
WIC-1T
WIC-2AM
WIC-2T
WIC-Cover
Physical Device View
Zoom In
Original Size
Zoom Out
Customize Icon in Physical View
Customize Icon in Logical View
The HWIC-2T is a Cisco 2-Port Serial High-Speed WAN Interface Card, providing 2 serial ports.


2911
Router1
2960-24TT
Switch0
2960-24TT
Switch1
PC-PT
PC0
PC-PT
PC1
PC-PT
PC2
PC-PT
PC3
PC-PT
PC4
PC-PT
PC5


Router>enable
Router#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface gigabitEthernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up
exit
Router(config)#interface gigabitEthernet 0/1
Router(config-if)#ip address 192.168.2.1 255.255.255.224
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
exit
Copy
Paste

ping 192.168.1.X

7. Test Packet Simulation:.

↳ Use simulation mode in the packet Tracer.

↳ Send a packet between PCs.

↳ Observe the packet movement through the switch and router

↳ Repeat the test without a router and observe the result.

OBSERVATION:.

↳ Devices within the same subnet can communicate directly.

↳ Different subnets require a router for communication.

↳ Packet transmission can be observed using simulation mode.

RESULT:-

Thus subnetting was successfully implemented in Cisco Packet Tracer, static IP addressing were assigned, and communication between different subnets was tested using a router.

PC0

Physical Config Desktop Custom Interface

IP Configuration

IP Configuration

DHCP Static

IP Address 192.168.1.11

Subnet Mask 255.255.255.0

Default Gateway 192.168.1.1

DNS Server

IPv6 Configuration

DHCP Auto Config Static

IPv6 Address /

Link Local Address FE80::2D0:BAFF:FE9C:D809

IPv6 Gateway

IPv6 DNS Server